

The Alaoglu–Erdős Integer-Exponent Conjecture

Continuation Report 28

Confluent Functional Calculus, Rational Self-Extensions,
and the Missing Centralizer Jet

New additive progress beyond the unified 27-report manuscript

August 22, 2026

Bottom line. The conjecture remains unproved in this report. The main advance is instead an exact classification theorem for every first-order rational self-extension of a semisimple matrix whose spectral values are already known. For a hypothetical nonintegral exponent x , the x -th power of such an extension is rational *exactly* in the split, commutator directions. Every genuinely non-split rational extension exposes the missing scalar x . Equivalently, real power acts on the rational self-extension quotient as x times a rational automorphism. This turns the informal slogan “a Jordan block would solve the problem” into a complete rationality-locus theorem.

A second advance develops a confluent approximation hierarchy. Symmetric transported secants improve the first-order error in the unified report from $O(h)$ to $O(h^2)$, and explicit Richardson coefficients raise this to $O(h^{2m})$ for arbitrary fixed m , while all approximants remain rational under the counterexample hypothesis. A denominator audit shows precisely why this still does not force rationality: the available heights are exponential in the Diophantine-approximation parameter, whereas the analytic gain is polynomial. The report also proves a toric-functor no-go theorem, gives a genuinely integral 3×3 Newton-chain example, and isolates an affine collision program based on the additive relation $3 = 2 + 1$.

Abstract

Assume that $x \in \mathbb{R}$ and

$$M := 2^x \in \mathbb{Z}, \quad N := 3^x \in \mathbb{Z}.$$

The attached unified manuscript establishes a large body of exact reductions, finite verification, determinant estimates, and conditional structure, while leaving the original conjecture open. This continuation deliberately avoids repeating that material. Its first purpose is to identify, in a basis-free way, the exact finite-dimensional datum absent from all semisimple constructions.

Let $A \in M_n(\mathbb{Q})$ be semisimple with positive rational eigenvalues $\lambda_1, \dots, \lambda_s$, rational spectral projectors, and $\lambda_i^x \in \mathbb{Q}$. For $H \in M_n(\mathbb{Q})$ form the first-order block extension

$$\mathcal{E}_A(H) = \begin{pmatrix} A & H \\ 0 & A \end{pmatrix}.$$

The analytic block functional calculus gives

$$\mathcal{E}_A(H)^x = \begin{pmatrix} A^x & D(z^x)_A[H] \\ 0 & A^x \end{pmatrix}.$$

If $x \notin \mathbb{Q}$, then

$$D(z^x)_A[H] \in M_n(\mathbb{Q}) \iff H \in [A, M_n(\mathbb{Q})].$$

Thus rationality survives exactly on extensions already split over $\mathbb{Q}$; every nonzero class in $M_n(\mathbb{Q})/[A, M_n(\mathbb{Q})]$ detects x . On that quotient the derivative is x times multiplication by the rational invertible matrix A^{x-1} . Several trace and matrix-coefficient extraction criteria follow.

The second purpose is to connect this exact obstruction with rational confluence. For $r = 2^a 3^b$ and $r^x = M^a N^b$, define

$$J_x(r) = \frac{r^x - r^{-x}}{r - r^{-1}} = \frac{\sinh(xh)}{\sinh h}, \quad h = \log r.$$

Then $J_x(r) \in \mathbb{Q}$ and $J_x(e^h) = x + O_x(h^2)$. With

$$c_{m,k} = (-1)^{k-1} \frac{2(m!)^2}{(m-k)!(m+k)!}, \quad R_{m,x}(r) = \sum_{k=1}^m c_{m,k} J_x(r^k),$$

we prove $R_{m,x}(e^h) = x + O_{x,m}(h^{2m})$. This is a genuine higher-order strengthening of the transported secants already present in the unified report, but the reduced-denominator problem remains decisive.

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1 Executive answer and evidentiary boundary

1.1 Outcome

The requested full proof was not obtained. What was obtained is a collection of exact new results that sharply narrows one previously qualitative obstruction:

1. **Complete first-order rationality locus.** For every rational semisimple base A with rational positive spectrum and rational powered spectrum, a hypothetical irrational exponent makes the rational block extensions with rational x -th power *exactly* the commutator, hence split, extensions.
2. **Extension-space action.** Modulo rational splitting, the Fréchet derivative of z^x is exactly x times a rational automorphism. This is the finite-dimensional location of the missing arithmetic datum.
3. **Toric no-go.** Direct sums, tensor products, duals, symmetric powers, exterior powers, Schur functors, and rational subquotients of the split torus representation generated by 2 and 3 can create multiplicity but never a nonzero extension class. Spectral collision is not confluence.
4. **Integral collision examples.** An explicit integral 3×3 Newton chain has integral x -th power under the original hypotheses, is not unimodularly diagonalizable, and produces repeated eigenvalues after a tensor square; nevertheless all collisions remain semisimple.
5. **Higher-order rational confluence.** A symmetric secant and its Richardson extrapolants yield rational approximations to x of arbitrary fixed even order. The exact denominator growth that blocks unit locking is isolated.
6. **Affine escape route.** The family $\begin{pmatrix} q & 1 \\ 0 & 3 \end{pmatrix}$, sampled at 3-smooth rational q , converges to the decisive Jordan extension at $q = 3$. Proving an arithmetic compactness statement for these rational secants would close the conjecture.

Status. The classification, quotient, toric no-go, explicit matrix formulas, degeneration theorem, Richardson identities, and conditional locking statements are [\[proved here\]](#) paper arguments in this report. They have been checked algebraically and by symbolic scripts reproduced in Appendix C, but they are not claimed to have been accepted by a proof assistant. The original Alaoglu–Erdős conjecture remains [\[open interface\]](#).

1.2 External reconnaissance as of August 22, 2026

Because the prompt specifically referred to recent public AI-assisted mathematics, the public record was checked separately from the mathematical derivation. Anthropic publicly launched Claude Fable 5 on June 9, 2026 and restored broad access on July 1 after a temporary suspension [9, 10]. The recent Jacobian development is a disproof in dimensions greater than two, with explicit constructions and exact verification; it does not settle the still-distinct two-dimensional case [11]. A public MathOverflow update and the ICARM leaderboard report a curve of Mordell–Weil rank at least 30, with exact rank 30 conditional on BSD and GRH [12, 13]. Public computational announcements also report exact constructions for the twelve previously open Hadamard orders

below 2000 [14, 15]; at the time of writing this is best treated as an exact-computation announcement rather than a conventional refereed paper.

No public source located in this search announced a proof of the Alaoglu–Erdős conjecture or a Lean formalization of one. The rival team’s reported result may therefore be private or unannounced. Nothing in this report assumes access to it.

1.3 The immovable transcendence boundary

The unified manuscript already explains that the conjecture is an integral special case of the four exponentials problem. The recent Matrix Coefficient Conjecture of Dasgupta and Kakde gives a particularly clean modern language for this boundary; its 2×2 case is equivalent to four exponentials [5]. No claim below silently assumes that conjecture. The point of the present work is instead to seek extra arithmetic structure coming from integrality, lattices, and confluence.

2 Non-duplication ledger

The attached manuscript is nearly one megabyte and already contains twenty-seven continuation reports. Repeating its reductions would obscure rather than advance the project. Table 1 records what is inherited and what is genuinely added here.

Table 1: Non-duplication ledger.

Topic	Already present in the unified manuscript	New contribution in Report 28
Basic reductions	Positivity, rational descent, transcendence of a hypothetical counterexample, exact conditional semigroup structure, finite verification	Used only as black-box baseline; not reproved except for two one-line closure lemmas
Four exponentials	Exact reduction and Matrix Coefficient formulation	No new transcendence claim; boundary is kept explicit
Semisimple matrices	Formula for a matrix satisfying $(T - 2I)(T - 3I) = 0$ and the observation that distinct-node functional calculus is blind	Full Fréchet derivative decomposition for every rational semisimple A , with an if-and-only-if rationality locus
Jordan blocks	A sufficient criterion from $(bI + N)^x = b^x I + xb^{x-1}N$	Classification of all first-order rational self-extensions, the extension quotient, and basis-free extraction maps
Tensor/symmetric constructions	Qualitative semisimple blindness	Categorical torus no-go theorem: multiplicity without extension, including rational subquotients

Topic	Already present in the unified manuscript	New contribution in Report 28
Transported secants	Rational one-sided secants with first-order error and finite-jet ceilings	Symmetric $O(h^2)$ secants and explicit Richardson extrapolation of order $2m$
Integral interpolation	Geometric Newton identities and Smith-form analyses	Explicit 3×3 lattice-glued chain at eigenvalues 2, 3, 4, plus tensor collision and unbounded-conjugator degeneration theorems
Operator/affine ideas	External affine ruler as a sufficient condition	A parameterized affine collision at $q = 3$ sampled entirely on the 3-smooth rational group
Lean plan	Many accepted modules and a prior Jordan closing lemma	A modular formalization plan for the new algebraic classification and Richardson identities

3 Baseline, notation, and the exact question

Assume throughout that

$$x \in \mathbb{R}, \quad M = 2^x \in \mathbb{Z}, \quad N = 3^x \in \mathbb{Z}. \quad (1)$$

If $x < 0$, then $0 < 2^x < 1$, impossible for an integer; hence $x \geq 0$. The case $x = 0$ is integral. In any hypothetical counterexample, therefore,

$$x > 0, \quad x \notin \mathbb{Q}, \quad M, N \geq 2. \quad (2)$$

The irrationality assertion is inherited: if $x = a/b \in \mathbb{Q}$ in lowest terms and $2^x \in \mathbb{Q}$, unique factorization forces $b = 1$. The unified report further proves that a nonintegral counterexample would be transcendental.

Let

$$\Gamma = \{2^a 3^b : a, b \in \mathbb{Z}\} \subset \mathbb{Q}_{>0}. \quad (3)$$

Then (1) gives an exact order-preserving group homomorphism

$$\rho_x : \Gamma \longrightarrow \mathbb{Q}_{>0}, \quad \rho_x(2^a 3^b) = M^a N^b = (2^a 3^b)^x. \quad (4)$$

Because $\log 2 / \log 3 \notin \mathbb{Q}$, the logarithms of elements of Γ form a dense additive subgroup of $\mathbb{R}$. In particular, there are sequences $r_j \in \Gamma \setminus \{1\}$ with $r_j \rightarrow 1$. This density creates confluence analytically; the entire problem is to make that confluence arithmetically rigid.

Definition 3.1 (Divided differences). For a function f on positive reals, write

$$f[\lambda, \mu] = \begin{cases} \frac{f(\lambda) - f(\mu)}{\lambda - \mu}, & \lambda \neq \mu, \\ f'(\lambda), & \lambda = \mu. \end{cases}$$

For $f_x(z) = z^x$, the diagonal value is $f_x[\lambda, \lambda] = x\lambda^{x-1}$.

The distinction between the off-diagonal and diagonal divided differences is the organizing principle of this report. If $\lambda, \mu, \lambda^x, \mu^x \in \mathbb{Q}$ and $\lambda \neq \mu$, then $f_x[\lambda, \mu] \in \mathbb{Q}$ automatically. At a repeated node, rationality of the divided difference is exactly rationality of $x\lambda^{x-1}$, hence—once λ^x is known—of x itself.

4 First-order block functional calculus

4.1 The universal block identity

For $A, H \in M_n(\mathbb{C})$ define

$$\mathcal{E}_A(H) = \begin{pmatrix} A & H \\ 0 & A \end{pmatrix}. \quad (5)$$

This is the universal first-order self-extension of A in the direction H .

Theorem 4.1 (Block derivative identity). *Let f be holomorphic on a neighborhood of the spectrum of A . Then*

$$f(\mathcal{E}_A(H)) = \begin{pmatrix} f(A) & Df_A[H] \\ 0 & f(A) \end{pmatrix}, \quad (6)$$

where Df_A is the Fréchet derivative of the holomorphic functional calculus at A .

Proof. For a positively oriented contour γ enclosing the spectrum,

$$f(B) = \frac{1}{2\pi i} \int_{\gamma} f(z)(zI - B)^{-1} dz.$$

The block inverse is exact:

$$(zI - \mathcal{E}_A(H))^{-1} = \begin{pmatrix} (zI - A)^{-1} & (zI - A)^{-1}H(zI - A)^{-1} \\ 0 & (zI - A)^{-1} \end{pmatrix}.$$

Integrating entrywise gives (6); the upper-right contour integral is the standard Fréchet derivative. $\square$

For the positive-spectrum matrices used below, $f_x(z) = z^x$ is taken on the principal branch near the spectrum, so Theorem 4.1 applies for every real x .

4.2 Spectral divided-difference formula

Theorem 4.2 (Semisimple derivative). *Let A be semisimple with distinct eigenvalues $\lambda_1, \dots, \lambda_s$ and spectral projectors $P_1, \dots, P_s$. Then*

$$Df_A[H] = \sum_{i,j=1}^s f[\lambda_i, \lambda_j] P_i H P_j. \quad (7)$$

In particular, for $f_x(z) = z^x$,

$$Df_{x,A}[H] = \sum_{i \neq j} \frac{\lambda_i^x - \lambda_j^x}{\lambda_i - \lambda_j} P_i H P_j + x \sum_i \lambda_i^{x-1} P_i H P_i. \quad (8)$$

Proof. Insert $(zI - A)^{-1} = \sum_i (z - \lambda_i)^{-1} P_i$ into the contour formula in the proof of Theorem 4.1. For $i \neq j$, partial fractions give the divided difference; for $i = j$, Cauchy's differentiation formula gives $f'(\lambda_i)$. $\square$

Equation (8) is already suggestive. Every off-diagonal coefficient is a secant and is rational from value data. Every diagonal block is multiplied by x . The next section turns that observation into an exact if-and-only-if theorem.

5 Exact rationality locus for first-order self-extensions

5.1 Centralizer and commutator directions

Let

$$\mathcal{Z}(A) = \{X \in M_n(\mathbb{Q}) : AX = XA\}$$

be the rational centralizer, and let

$$[A, M_n(\mathbb{Q})] = \{AY - YA : Y \in M_n(\mathbb{Q})\}$$

be the image of the commutator map. Under the rational semisimplicity hypotheses below, these are complementary subspaces.

Lemma 5.1 (Rational centralizer–commutator splitting). *Suppose $A \in M_n(\mathbb{Q})$ is semisimple, all its eigenvalues $\lambda_1, \dots, \lambda_s$ lie in $\mathbb{Q}$, and its spectral projectors P_i lie in $M_n(\mathbb{Q})$. Then*

$$M_n(\mathbb{Q}) = \mathcal{Z}(A) \oplus [A, M_n(\mathbb{Q})]. \quad (9)$$

The two projections are

$$H_c = \sum_i P_i H P_i, \quad H_o = \sum_{i \neq j} P_i H P_j. \quad (10)$$

Moreover,

$$H_o = [A, Y_H], \quad Y_H = \sum_{i \neq j} \frac{1}{\lambda_i - \lambda_j} P_i H P_j. \quad (11)$$

Proof. The block decomposition $H = \sum_{i,j} P_i H P_j$ is rational. Its diagonal part commutes with A . For $i \neq j$,

$$[A, P_i H P_j] = (\lambda_i - \lambda_j) P_i H P_j,$$

which proves (11). Finally, a commutator has zero diagonal spectral blocks, so the intersection of the two summands is zero. $\square$

Lemma 5.2 (Commutator covariance). *For every holomorphic f and every Y ,*

$$Df_A[[A, Y]] = [f(A), Y]. \quad (12)$$

Proof. Differentiate at $t = 0$ the similarity identity

$$f(e^{-tY} A e^{tY}) = e^{-tY} f(A) e^{tY}.$$

Equivalently, verify the identity termwise in (7). $\square$

This covariance explains why commutator directions carry no new information: they are already forced by ordinary similarity of the known value $f(A)$.

5.2 The classification theorem

Theorem 5.3 (Exact first-order rationality locus). *Let $A \in M_n(\mathbb{Q})$ be semisimple with positive rational eigenvalues $\lambda_1, \dots, \lambda_s$ and rational spectral projectors. Let $x \in \mathbb{R} \setminus \mathbb{Q}$ and assume*

$$\lambda_i^x \in \mathbb{Q} \quad (1 \leq i \leq s). \quad (13)$$

Then for every $H \in M_n(\mathbb{Q})$,

$$Df_{x,A}[H] \in M_n(\mathbb{Q}) \iff H \in [A, M_n(\mathbb{Q})]. \quad (14)$$

Equivalently,

$$\mathcal{E}_A(H)^x \in M_{2n}(\mathbb{Q}) \iff \mathcal{E}_A(H) \text{ is rationally similar to } A \oplus A. \quad (15)$$

Proof. Write $H = H_c + H_o$ as in Lemma 5.1. The off-diagonal part of (8) is rational because each coefficient

$$\frac{\lambda_i^x - \lambda_j^x}{\lambda_i - \lambda_j}$$

is rational. The centralizer part is

$$Df_{x,A}[H_c] = x \sum_i \lambda_i^{x-1} P_i H P_i = x C_H, \quad (16)$$

where $C_H \in M_n(\mathbb{Q})$. If $Df_{x,A}[H]$ is rational, then $x C_H$ is rational. Since x is irrational, every rational entry of C_H must vanish, so $C_H = 0$. Multiplying on the left and right by P_i gives

$$\lambda_i^{x-1} P_i H P_i = 0.$$

The scalar is nonzero, hence every diagonal spectral block vanishes. Thus $H = H_o \in [A, M_n(\mathbb{Q})]$.

Conversely, if $H = [A, Y]$, Lemma 5.2 gives

$$Df_{x,A}[H] = [A^x, Y] \in M_n(\mathbb{Q}),$$

because $A^x = \sum_i \lambda_i^x P_i$ is rational.

For the similarity statement, if $H = [A, Y]$, then

$$\mathcal{E}_A(H) = \begin{pmatrix} I & -Y \\ 0 & I \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix} \begin{pmatrix} I & Y \\ 0 & I \end{pmatrix}. \quad (17)$$

Conversely, write $H = H_c + [A, Y_H]$ by Lemma 5.1. The same block-unipotent calculation conjugates $\mathcal{E}_A(H)$ to $\mathcal{E}_A(H_c)$. On the λ_i -primary summand of the latter, the nilpotent part is $\begin{pmatrix} 0 & P_i H P_i \\ 0 & 0 \end{pmatrix}$. If $\mathcal{E}_A(H)$ is similar to the semisimple matrix $A \oplus A$, every such nilpotent part must vanish. Hence $H_c = 0$, so H is a commutator. $\square$

Corollary 5.4 (A single non-split extension closes Alaoglu–Erdős). *Assume (1). Let A satisfy the hypotheses of Theorem 5.3, with all eigenvalues in Γ , and let $H \in M_n(\mathbb{Q})$ have a nonzero centralizer component. If*

$$\mathcal{E}_A(H)^x \in M_{2n}(\mathbb{Q}),$$

then $x \in \mathbb{Z}$.

Proof. Every eigenvalue $\lambda_i = 2^{a_i} 3^{b_i}$ has $\lambda_i^x = M^{a_i} N^{b_i} \in \mathbb{Q}$. If x were irrational, Theorem 5.3 would force the centralizer component of H to vanish. Thus $x \in \mathbb{Q}$, and rational descent from $2^x \in \mathbb{Q}$ gives $x \in \mathbb{Z}$. $\square$

Remark 5.5. The prior Jordan criterion corresponds to the special case $A = [b]$ and $H = [1]$. Theorem 5.3 is stronger in two ways: it treats arbitrary rational semisimple bases and it proves necessity as well as sufficiency. It identifies all false positives: they are exactly the rationally split extensions.

6 The extension quotient and the missing scalar

6.1 Self-extension space

Define the first-order rational self-extension quotient

$$\mathrm{Ext}_A^{1,\mathrm{lin}} := \mathrm{M}_n(\mathbb{Q}) / [A, \mathrm{M}_n(\mathbb{Q})]. \quad (18)$$

This notation is intentionally elementary: no derived-category machinery is needed. Two block extensions $\mathcal{E}_A(H)$ and $\mathcal{E}_A(H')$ are identified when a rational block-unipotent change of basis changes one into the other, equivalently when $H - H'$ is a commutator.

If A has spectral multiplicities $m_1, \dots, m_s$, then

$$\dim_{\mathbb{Q}} \mathrm{Ext}_A^{1,\mathrm{lin}} = \sum_{i=1}^s m_i^2, \quad (19)$$

because the quotient is canonically represented by the block centralizer $\bigoplus_i \mathrm{End}_{\mathbb{Q}}(P_i \mathbb{Q}^n)$.

Theorem 6.1 (Power acts as x times a rational automorphism). *Under the hypotheses of Theorem 5.3, the derivative induces a map*

$$\overline{Df}_{x,A} : \mathrm{Ext}_A^{1,\mathrm{lin}} \longrightarrow \mathrm{Ext}_{A^x}^{1,\mathrm{lin}}. \quad (20)$$

The centralizers and commutator spaces of A and A^x coincide, and after the natural identification of the two quotients,

$$\overline{Df}_{x,A}([H]) = x [A^{x-1} H]. \quad (21)$$

Multiplication by $A^{x-1} = A^x A^{-1} \in \mathrm{M}_n(\mathbb{Q})$ is an invertible rational linear map on the quotient. Hence the only non-rational scalar in the induced action is x .

Proof. Lemma 5.2 shows that commutators map to commutators, so the map is well-defined. Since $x > 0$ in the substantive case, the map $t \mapsto t^x$ is injective on the positive spectrum; A and A^x therefore have the same spectral projectors and the same block-diagonal/off-diagonal decomposition. Modulo the off-diagonal commutator space, (8) becomes

$$[Df_{x,A}[H]] = x[A^{x-1}H].$$

The matrix $A^{x-1} = A^x A^{-1}$ is rational and invertible. Left multiplication by it preserves each spectral block and the off-diagonal subspace, so it induces a rational automorphism of the quotient. $\square$

The original scalar problem can therefore be restated as follows. The hypotheses give the value-level map $A \mapsto A^x$ on every semisimple rational representation of the 3-smooth torus. To prove $x \in \mathbb{Z}$, it is enough to show that this value-level map preserves one nonzero rational first-order self-extension class. On every such class, the induced scalar is x .

6.2 Basis-free extraction by traces and matrix coefficients

The quotient theorem gives several concrete extraction formulas. They are useful both for paper arguments and for formalization because they avoid choosing eigenvectors.

Proposition 6.2 (Weighted trace identity). *Under the hypotheses of Theorem 6.1, for every integer $m \geq 0$ and every $H \in M_n(\mathbb{Q})$,*

$$\mathrm{tr}(A^{m+1-x} Df_{x,A}[H]) = x \mathrm{tr}(A^m H). \quad (22)$$

Here

$$A^{m+1-x} = A^{m+1}(A^x)^{-1} \in M_n(\mathbb{Q}).$$

Proof. Only diagonal spectral blocks contribute to a trace. On the i -th diagonal block, $Df_{x,A}$ acts by $x\lambda_i^{x-1}$, while A^{m+1-x} acts by λ_i^{m+1-x} . Their product is $x\lambda_i^m$. Summing the block traces gives (22). $\square$

Corollary 6.3 (Moment extraction). *If $Df_{x,A}[H]$ is rational and $\mathrm{tr}(A^m H) \neq 0$ for some $m \geq 0$, then $x \in \mathbb{Q}$. Under (1), this implies $x \in \mathbb{Z}$.*

Proof. The left side of (22) is rational and the right side is a nonzero rational multiple of x . $\square$

Proposition 6.4 (Finite detection for simple spectrum). *Suppose in addition that A has n distinct rational eigenvalues. If the diagonal spectral component of H is nonzero, then*

$$\mathrm{tr}(A^m H) \neq 0$$

for at least one $m \in \{0, 1, \dots, n-1\}$.

Proof. Choose a rational eigenbasis. Let h_i be the i -th diagonal entry of H . The equations $\sum_i \lambda_i^m h_i = 0$ for $0 \leq m < n$ have Vandermonde coefficient matrix and therefore force every $h_i = 0$. $\square$

Caution 6.5. With spectral multiplicities, traces detect only the traces of the centralizer blocks; a nonzero traceless block may escape all scalar moments. The full theorem is nevertheless recovered by rational matrix coefficients $u^T P_i Df_{x,A}[H] P_i v$, or simply by the block projection itself. No trace-only claim is made in that generality.

7 The complete 2–3 block model

Take

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad H = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{Q}). \quad (23)$$

Then $A^x = \text{diag}(M, N)$, and Theorem 4.2 gives the exact formula

$$Df_{x,A}[H] = \begin{pmatrix} \frac{xM}{2}a & (N-M)b \\ (N-M)c & \frac{xN}{3}d \end{pmatrix}. \quad (24)$$

Consequently

$$\begin{pmatrix} 2 & 0 & a & b \\ 0 & 3 & c & d \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}^x = \begin{pmatrix} M & 0 & \frac{xM}{2}a & (N-M)b \\ 0 & N & (N-M)c & \frac{xN}{3}d \\ 0 & 0 & M & 0 \\ 0 & 0 & 0 & N \end{pmatrix}. \quad (25)$$

Theorem 7.1 (Explicit rationality test). *Assume (1) and $x \notin \mathbb{Q}$. For H as in (23),*

$$\mathcal{E}_A(H)^x \in M_4(\mathbb{Q}) \iff a = d = 0. \quad (26)$$

The matrices with $a = d = 0$ are precisely the commutators $[A, Y]$.

Proof. The off-diagonal entries in (24) are rational. Since M, N are nonzero rational numbers and x is irrational, the diagonal entries are rational precisely when $a = d = 0$. Directly,

$$[A, \begin{pmatrix} u & v \\ w & z \end{pmatrix}] = \begin{pmatrix} 0 & -v \\ w & 0 \end{pmatrix},$$

which is exactly the off-diagonal subspace. □

Corollary 7.2 (Minimal concrete closing target). *It would suffice to derive from the original scalar hypotheses the rationality of the x -th power of any one of the following matrices:*

$$\begin{pmatrix} 2 & 0 & 1 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

The exposed entry is respectively $xM/2$ or $xN/3$, and rational descent then gives $x \in \mathbb{Z}$.

7.1 False positives are exactly split directions

For any $u, v \in \mathbb{Q}$, the off-diagonal direction $H = \begin{pmatrix} 0 & u \\ v & 0 \end{pmatrix}$ produces a rational powered block. This is not partial evidence for a derivative. Equation (17) shows that it is just a change of rational basis of $A \oplus A$. The classification theorem is therefore also a debugging tool: any proposed finite-dimensional construction should first be projected to the quotient (18). If its class vanishes, no amount of entrywise complexity can reveal x .

8 Why toric functors create multiplicity but not confluence

8.1 Weight-space formulation

Let $T = \mathbb{G}_m^2$ over $\mathbb{Q}$. A finite-dimensional rational representation of T splits as a direct sum of weight spaces

$$V = \bigoplus_{(a,b) \in \mathbb{Z}^2} V_{a,b}, \quad (s, t) \cdot v = s^a t^b v \quad (v \in V_{a,b}). \quad (27)$$

This is not merely diagonalizability at one matrix: a split torus is linearly reductive in characteristic zero, so every rational subrepresentation and every quotient again decomposes into weights.

Evaluating at $(2, 3)$ gives eigenvalue $2^a 3^b$ on $V_{a,b}$; evaluating at (M, N) gives $M^a N^b = (2^a 3^b)^x$. Unique factorization implies

$$2^a 3^b = 2^{a'} 3^{b'} \iff (a, b) = (a', b'). \quad (28)$$

The same remains true after applying $x > 0$.

Theorem 8.1 (Toric-functor no-go). *Assume $x > 0$. Start with finite-dimensional rational T -representations and apply any finite combination of*

1. *direct sums and tensor products;*
2. *duals;*
3. *symmetric, exterior, or more general Schur functors;*
4. *rational T -stable subspaces and rational quotients.*

After evaluation at $(2, 3)$ or at (M, N) , the resulting operator is semisimple. Repeated eigenvalues can arise only from repeated occurrences of the same character; the operator on each repeated eigenspace is scalar. In particular, every self-extension arising within the toric representation category is split and maps to the zero class in (18).

Proof. Every listed operation stays inside the semisimple tensor category of rational representations of the split torus. Thus the output is a direct sum of character spaces. Evaluation sends a character (a, b) to the positive scalar $2^a 3^b$ or $M^a N^b$. By (28), distinct characters do not collide under either evaluation. Multiplicity can therefore only mean that one character occurs several times, and the torus element acts as a scalar on that entire multiplicity space. In particular, every self-extension constructed inside this toric representation category splits; its image in $\text{Ext}_A^{1, \text{lin}}$ is the zero class. (The ambient quotient $\text{Ext}_A^{1, \text{lin}}$ itself is generally nonzero.) $\square$

Corollary 8.2 (Spectral collision is not confluence). *Tensor-product identities such as*

$$(2)(3) = (3)(2)$$

produce a repeated eigenvalue 6, but the two copies are copies of the same weight and remain a direct sum. They do not produce the derivative coefficient $x6^{x-1}$.

Remark 8.3. This theorem does not say that tensor constructions are useless for every purpose. They can amplify valuations, create determinant identities, and expose multiplicities. It says something narrower and exact: without a non-semisimple operation or a singular limiting process, they cannot cross from value data to first-jet data.

8.2 Where a successful construction must leave the torus category

There are three mathematically distinct exits:

An actual extension. Produce a rational matrix with a nonzero class in $\text{Ext}_A^{1,\text{lin}}$ and prove its x -th power rational.

A singular degeneration. Approach a repeated spectral point through distinct rational spectral points while controlling denominators strongly enough that the limiting jet is arithmetic.

An affine relation. Add translation or another non-toric operation. The relation $3 = 2 + 1$ is the smallest available additive input.

The remainder of the report studies the last two exits.

9 Integral lattice gluing without a jet

9.1 A 3×3 Newton chain at 2, 3, 4

Consider the integral upper-bidiagonal matrix

$$T = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{pmatrix}. \quad (29)$$

Its eigenvalues are distinct, so it is semisimple over $\mathbb{Q}$, but its integral lattice is nontrivially glued across the three eigenspaces.

Theorem 9.1 (Integral Newton-chain power). *Under (1),*

$$T^x = \begin{pmatrix} M & N - M & \frac{M^2 - 2N + M}{2} \\ 0 & N & M^2 - N \\ 0 & 0 & M^2 \end{pmatrix} \in M_3(\mathbb{Z}). \quad (30)$$

Proof. For an upper-bidiagonal matrix with distinct diagonal entries, functional calculus gives first divided differences on the first superdiagonal and the second divided difference on the second superdiagonal. Here

$$\begin{aligned} f_x[2, 3] &= N - M, \\ f_x[3, 4] &= M^2 - N, \\ f_x[2, 3, 4] &= \frac{f_x[3, 4] - f_x[2, 3]}{4 - 2} = \frac{M^2 - 2N + M}{2}. \end{aligned}$$

The final numerator is even because $M^2 + M$ is even and $2N$ is even. Thus every entry is an integer. $\square$

Proposition 9.2 (The lattice gluing is genuinely integral). *The matrix T is diagonalizable over $\mathbb{Q}$ but not conjugate to $\text{diag}(2, 3, 4)$ by an element of $\text{GL}_3(\mathbb{Z})$. The lattice generated by primitive integral eigenvectors has index 2 in $\mathbb{Z}^3$.*

Proof. Primitive eigenvectors for 2, 3, 4 are respectively

$$v_2 = (1, 0, 0)^T, \quad v_3 = (1, 1, 0)^T, \quad v_4 = (1, 2, 2)^T.$$

Their column determinant is 2. Any integral eigenbasis has the same primitive lines and therefore determinant divisible by 2, so no unimodular eigenbasis exists. $\square$

This example matters because it rules out an overly optimistic inference: “if a rational semisimple matrix is integrally glued, then integrality of its irrational power should force an integer exponent.” The original hypotheses themselves make (30) integral. Integral gluing is not yet a self-extension.

9.2 A concrete repeated-spectrum false positive

The tensor square $T \otimes T$ is integral and semisimple. Its eigenvalue multiset is

$$4, 6, 8, 6, 9, 12, 8, 12, 16,$$

so 6, 8, 12 occur with multiplicity. Moreover,

$$(T \otimes T)^x = T^x \otimes T^x \in M_9(\mathbb{Z}). \quad (31)$$

The equality follows after rational diagonalization, since the eigenvalues of a tensor product are products and $(\lambda\mu)^x = \lambda^x\mu^x$ for positive scalars.

Audit finding 9.3 (Collision without confluence). The matrix $T \otimes T$ has repeated eigenvalues, a nontrivial integral lattice, and an integral x -th power. Nevertheless it is semisimple, and every repeated eigenspace carries a scalar operator. Its first-order extension class is zero. Thus none of the three visible features — repeated spectrum, integral gluing, or integral powered entries — substitutes for a Jordan jet.

10 Split matrices can degenerate to a Jordan block

The previous no-go theorem is exact at every finite stage, but semisimplicity is not closed under singularly changing bases. This produces a useful model of the analytic limit and a warning about arithmetic compactness. Such integral collisions exist: if p_j/q_j are continued-fraction convergents to $\log 3/\log 2$, then one may take $a_j = 2^{p_j}$ and $b_j = 3^{q_j}$ along either side of the approximation.

Theorem 10.1 (Unbounded integral conjugators recover the first jet). *Let a_j, b_j be positive elements of the integral 3-smooth semigroup $\{2^u 3^v : u, v \in \mathbb{N}_0\}$ such that*

$$r_j := \frac{b_j}{a_j} \longrightarrow 1, \quad b_j \neq a_j. \quad (32)$$

Choose integers k_j satisfying

$$k_j(r_j - 1) \longrightarrow 1. \quad (33)$$

For example, take an integer nearest to $(r_j - 1)^{-1}$. Define

$$S_j = \begin{pmatrix} 1 & -k_j \\ 0 & 1 \end{pmatrix}, \quad B_j = S_j^{-1} \begin{pmatrix} a_j & 0 \\ 0 & b_j \end{pmatrix} S_j. \quad (34)$$

Then $B_j \in M_2(\mathbb{Z})$, $B_j^x \in M_2(\mathbb{Z})$, and

$$a_j^{-1} B_j \longrightarrow J := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad (35)$$

$$a_j^{-x} B_j^x \longrightarrow J^x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}. \quad (36)$$

Proof. A direct multiplication gives

$$B_j = \begin{pmatrix} a_j & k_j(b_j - a_j) \\ 0 & b_j \end{pmatrix}. \quad (37)$$

It is integral and integrally similar to a diagonal matrix. Since a_j^x and b_j^x are integers,

$$B_j^x = \begin{pmatrix} a_j^x & k_j(b_j^x - a_j^x) \\ 0 & b_j^x \end{pmatrix} \in M_2(\mathbb{Z}). \quad (38)$$

After division by a_j , the upper-right entry of (37) is $k_j(r_j - 1)$, which tends to 1 by (33). For the power, the normalized upper-right entry is

$$k_j(r_j^x - 1) = k_j(r_j - 1) \frac{r_j^x - 1}{r_j - 1} \longrightarrow x.$$

The diagonal entries converge to 1, proving both limits. $\square$

Remark 10.2. The theorem is an exact matrix avatar of transported secants. At every finite stage the extension is split by the integral matrix S_j , but $\|S_j\| \rightarrow \infty$. The limit is a non-split Jordan block, and its powered limit contains x . The obstruction is therefore not analytic existence of the jet; it is the lack of a bounded arithmetic lattice in which the limit can be taken.

Corollary 10.3 (Symmetric powers expose the full limiting jet tower). *For each fixed $d \geq 1$, apply the d -th symmetric power to Theorem 10.1. Then*

$$a_j^{-dx} (\text{Sym}^d(B_j))^x = a_j^{-dx} \text{Sym}^d(B_j^x) = \text{Sym}^d(a_j^{-x} B_j^x) \longrightarrow \text{Sym}^d \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

In the monomial basis, the limiting entries are integer binomial multiples of $1, x, x^2, \dots, x^d$.

Proof. The identity holds after diagonalizing B_j : both sides have the same eigenvectors and weights $(a_j^{d-k} b_j^k)^x$. Symmetric power is polynomial in the matrix entries, so it commutes with the displayed limits. $\square$

Caution 10.4. Corollary 10.3 does not contradict the toric no-go theorem. Every finite j remains semisimple and split. The higher jets appear only after an unbounded change of basis and non-integral normalizations by powers of a_j^x . The denominators escaping through those normalizations are exactly the arithmetic difficulty.

11 Affine collision at the additive relation $3 = 2 + 1$

The torus category knows multiplication but not addition. The smallest way to leave it while remaining close to the original bases is to use the additive relation $3 = 2 + 1$.

11.1 A rational secant family converging to the decisive Jordan block

For $q \in \mathbb{Q}_{>0}$ define

$$C(q) = \begin{pmatrix} q & 1 \\ 0 & 3 \end{pmatrix}. \quad (39)$$

For $q \neq 3$, the eigenvalues are distinct and

$$C(q)^x = \begin{pmatrix} q^x & \frac{q^x - N}{q - 3} \\ 0 & N \end{pmatrix}. \quad (40)$$

At $q = 3$, the removable divided difference becomes a derivative:

$$C(3)^x = \begin{pmatrix} N & x3^{x-1} \\ 0 & N \end{pmatrix}. \quad (41)$$

If $q \in \Gamma$, then $q^x = \rho_x(q) \in \mathbb{Q}$, so every entry of (40) is rational. Since Γ is dense in logarithmic scale, there are $q_j \in \Gamma \setminus \{3\}$ with $q_j \rightarrow 3$. Thus the decisive irrationality detector $x3^{x-1}$ is the real limit of rational matrix entries supplied entirely by the original value data.

Proposition 11.1 (Affine collision criterion). *Assume (1). Any one of the following statements implies $x \in \mathbb{Z}$:*

1. $C(3)^x \in M_2(\mathbb{Q})$;
2. the rational secants $(q^x - N)/(q - 3)$, for $q \in \Gamma$ approaching 3, are eventually constant;
3. these secants lie in a fixed discrete subgroup $D^{-1}\mathbb{Z} \subset \mathbb{Q}$ in a neighborhood of 3;
4. their reduced denominators remain bounded along some sequence $q_j \rightarrow 3$.

Proof. The first statement makes $x3^{x-1}$ rational, hence $x = 3(x3^{x-1})/N \in \mathbb{Q}$. For the others, the secants converge in $\mathbb{R}$ to $x3^{x-1}$. An eventually constant rational sequence has a rational limit. A convergent sequence in a fixed discrete subgroup is eventually constant. Reduced denominators bounded by D put all terms in $(\text{lcm}(1, \dots, D))^{-1}\mathbb{Z}$, reducing to the discrete-subgroup case. Rational descent finishes. $\square$

Research target 11.2 (Arithmetic compactness at an affine collision). Prove any denominator bound, congruence stabilization, adelic compactness principle, or exact lattice realization strong enough to trigger Proposition 11.1 for $q \in \Gamma$ approaching 3.

This target is genuinely non-toric. The parameter q moves through a multiplicative group, but the collision point is selected by the affine difference $q - 3$. The denominator $q - 3$ is where additive arithmetic enters.

11.2 Integral orders and the conductor defect

The secant denominator has an exact lattice interpretation. Let $a, b, A, B \in \mathbb{Z}$ with $a \neq b$. For an integer $d \geq 1$, define the order

$$\mathcal{O}_d = \{(u, v) \in \mathbb{Z}^2 : u \equiv v \pmod{d}\} \subset \mathbb{Q} \times \mathbb{Q}. \quad (42)$$

Multiplication by (a, b) preserves $\mathcal{O}_d$ exactly when $d \mid a - b$. In the basis $(1, 1), (0, d)$ it has matrix

$$\begin{pmatrix} a & 0 \\ (b-a)/d & b \end{pmatrix}. \quad (43)$$

Multiplication by (A, B) preserves the same order exactly when $d \mid A - B$.

Definition 11.3 (Common conductor and conductor defect). For a pair of integral 3-smooth numbers a, b , put

$$a_x = a^x \in \mathbb{Z}, \quad b_x = b^x \in \mathbb{Z}, \quad g_x(a, b) = \gcd(a - b, a_x - b_x), \quad (44)$$

and define

$$\delta_x(a, b) = \frac{|a - b|}{g_x(a, b)}. \quad (45)$$

The integer $g_x(a, b)$ is the largest conductor d for which both the input element (a, b) and its powered output (a_x, b_x) preserve the same glued order $\mathcal{O}_d$. The defect $\delta_x(a, b)$ is exactly the reduced denominator of the ordinary divided difference $(a_x - b_x)/(a - b)$.

Proposition 11.4 (Order-preservation reformulation). *For integral 3-smooth $a \neq b$, the following are equivalent:*

1. $(a^x - b^x)/(a - b) \in \mathbb{Z}$;
2. $a - b \mid a^x - b^x$;
3. the maximal input order $\mathcal{O}_{|a-b|}$ is also preserved by (a^x, b^x) ;
4. the matrix $\begin{pmatrix} a & 1 \\ 0 & b \end{pmatrix}^x$ is integral.

More generally, the reduced denominator in (1) is $\delta_x(a, b)$.

Proof. The equivalence of the first three statements follows immediately from the order calculation. The fourth follows from the two-point divided-difference formula. Reducing the fraction $(a^x - b^x)/(a - b)$ gives denominator $|a - b|/\gcd(a - b, a^x - b^x)$. $\square$

Remark 11.5. This formulation suggests a concrete computational reconnaissance program: search close 3-smooth pairs not merely for a small real gap, but for anomalously small conductor defect. The real approximation and the lattice preservation must occur simultaneously. A small gap without common conductor reproduces the old denominator wall; a large common conductor is the arithmetic signature of genuine confluence.

12 Symmetric transported secants

The transported secants in the unified report use

$$\frac{r^x - 1}{r - 1} = \frac{e^{xh} - 1}{e^h - 1}, \quad h = \log r,$$

and approximate x with a linear error in h . A symmetric quotient cancels every odd error term automatically.

Definition 12.1 (Symmetric secant). For $r \in \Gamma \setminus \{1\}$ define

$$J_x(r) = \frac{r^x - r^{-x}}{r - r^{-1}}. \quad (46)$$

Writing $r = e^h$ gives

$$J_x(e^h) = \frac{\sinh(xh)}{\sinh h}. \quad (47)$$

Under (1), $r^x \in \mathbb{Q}$ and hence $J_x(r) \in \mathbb{Q}$.

Theorem 12.2 (Second-order rational confluence). For fixed real x ,

$$J_x(e^h) = x + \frac{x(x^2 - 1)}{6}h^2 + \frac{x(3x^4 - 10x^2 + 7)}{360}h^4 + O_x(h^6) \quad (h \rightarrow 0). \quad (48)$$

In particular,

$$J_x(e^h) = x + O_x(h^2). \quad (49)$$

Proof. Divide the Taylor series

$$\sinh(xh) = xh + \frac{x^3h^3}{6} + \frac{x^5h^5}{120} + O_x(h^7)$$

by

$$\sinh h = h + \frac{h^3}{6} + \frac{h^5}{120} + O(h^7).$$

Formal series division gives (48). Evenness in h also shows directly that all odd powers vanish. $\square$

Remark 12.3. The coefficient of h^4 is included because it is a useful audit check. For $x = 2$, $J_2(e^h) = 2 \cosh h = 2 + h^2 + h^4/12 + \dots$, and the formula gives $1/12$. For $x = 1$, every correction vanishes and $J_1 \equiv 1$.

12.1 Exact arithmetic formula

Let $r = B/A$ and $r^x = D/C$, where A, B, C, D are positive integers. No coprimality assumption is needed for the following unreduced identity:

$$J_x(r^k) = \frac{A^k B^k (D^{2k} - C^{2k})}{C^k D^k (B^{2k} - A^{2k})}. \quad (50)$$

For the canonical near-collisions $r = 3^q/2^p$, one may take

$$A = 2^p, \quad B = 3^q, \quad C = M^p, \quad D = N^q.$$

Formula (50) makes the height problem visible: even when $|\log r|$ is of order $1/q$, the unreduced numerator and denominator have exponential size in $p + q$.

13 Richardson extrapolation to arbitrary even order

13.1 The coefficients

For integers $m \geq 1$ and $1 \leq k \leq m$, define

$$c_{m,k} = (-1)^{k-1} \frac{2(m!)^2}{(m-k)!(m+k)!}. \quad (51)$$

These are the Lagrange coefficients for evaluating at 0 a polynomial given at the nodes $1^2, 2^2, \dots, m^2$.

Lemma 13.1 (Moment cancellation). *For every $m \geq 1$,*

$$\sum_{k=1}^m c_{m,k} = 1, \quad (52)$$

$$\sum_{k=1}^m c_{m,k} k^{2j} = 0 \quad (1 \leq j < m). \quad (53)$$

Proof. Let $L_k(t)$ be the Lagrange basis polynomial for the nodes $1^2, \dots, m^2$. Its value at zero is

$$\begin{aligned} L_k(0) &= \prod_{\substack{1 \leq \ell \leq m \\ \ell \neq k}} \frac{-\ell^2}{k^2 - \ell^2} \\ &= (-1)^{k-1} \frac{2(m!)^2}{(m-k)!(m+k)!} = c_{m,k}. \end{aligned}$$

Applying the interpolation identity $P(0) = \sum_k L_k(0)P(k^2)$ to $P(t) = 1$ gives (52); applying it to $P(t) = t^j$, $1 \leq j < m$, gives (53). $\square$

The first rows are

$$\begin{aligned} (c_{1,k}) &= (1), \\ (c_{2,k}) &= \left(\frac{4}{3}, -\frac{1}{3}\right), \\ (c_{3,k}) &= \left(\frac{3}{2}, -\frac{3}{5}, \frac{1}{10}\right), \\ (c_{4,k}) &= \left(\frac{8}{5}, -\frac{4}{5}, \frac{8}{35}, -\frac{1}{35}\right). \end{aligned}$$

13.2 The extrapolants

Definition 13.2 (Order- $2m$ rational jet approximant). For $r \in \Gamma \setminus \{1\}$, define

$$R_{m,x}(r) = \sum_{k=1}^m c_{m,k} J_x(r^k). \quad (54)$$

Every term is rational under (1), so

$$R_{m,x}(r) \in \mathbb{Q}. \quad (55)$$

Explicitly,

$$R_{1,x}(r) = J_x(r), \quad (56)$$

$$R_{2,x}(r) = \frac{4J_x(r) - J_x(r^2)}{3}, \quad (57)$$

$$R_{3,x}(r) = \frac{15J_x(r) - 6J_x(r^2) + J_x(r^3)}{10}, \quad (58)$$

$$R_{4,x}(r) = \frac{56J_x(r) - 28J_x(r^2) + 8J_x(r^3) - J_x(r^4)}{35}. \quad (59)$$

Theorem 13.3 (Arbitrary fixed even-order confluence). *For every fixed $x \in \mathbb{R}$ and $m \geq 1$,*

$$R_{m,x}(e^h) = x + O_{x,m}(h^{2m}) \quad (h \rightarrow 0). \quad (60)$$

More precisely, if

$$\frac{\sinh(xz)}{\sinh z} = x + \sum_{j \geq 1} a_j(x) z^{2j} \quad (61)$$

near $z = 0$, then

$$R_{m,x}(e^h) - x = a_m(x) h^{2m} \sum_{k=1}^m c_{m,k} k^{2m} + O_{x,m}(h^{2m+2}). \quad (62)$$

Proof. Insert (61) with $z = kh$ into (54). The constant term is $x \sum_k c_{m,k} = x$. For $1 \leq j < m$, the coefficient of h^{2j} is $a_j(x) \sum_k c_{m,k} k^{2j} = 0$ by Lemma 13.1. The first term not forced to vanish is $j = m$, which gives (62) and hence (60). $\square$

Proposition 13.4 (First surviving moment). *The first uncanceled moment is*

$$\sum_{k=1}^m c_{m,k} k^{2m} = (-1)^{m-1} (m!)^2. \quad (63)$$

Proof. Apply the degree- m Lagrange interpolation remainder to $P(t) = t^m$ at $t = 0$. Since the interpolating polynomial through the nodes $1^2, \dots, m^2$ has degree at most $m-1$,

$$P(0) - \sum_k c_{m,k} P(k^2) = \prod_{k=1}^m (0 - k^2) = (-1)^m (m!)^2.$$

As $P(0) = 0$, rearrangement gives (63). $\square$

Combining Theorem 13.3 and Proposition 13.4 gives the sharpened leading term

$$R_{m,x}(e^h) - x = (-1)^{m-1} (m!)^2 a_m(x) h^{2m} + O_{x,m}(h^{2m+2}). \quad (64)$$

13.3 Existence of rational approximants

By continued fractions for $\log 3 / \log 2$, there are infinitely many coprime positive pairs (p, q) with

$$h_{p,q} = q \log 3 - p \log 2 \longrightarrow 0. \quad (65)$$

Taking

$$r_{p,q} = \frac{3^q}{2^p} \in \Gamma$$

and applying Theorem 13.3 yields the following.

Corollary 13.5 (Rational approximations of every fixed even order). *Under (1), for every fixed $m \geq 1$ there are explicitly computable rational numbers*

$$R_{m,x} \left(\frac{3^q}{2^p} \right) \quad (66)$$

with

$$\left| R_{m,x} \left(\frac{3^q}{2^p} \right) - x \right| \ll_{x,m} |q \log 3 - p \log 2|^{2m}. \quad (67)$$

For convergents p/q , the right side is $O_{x,m}(q^{-2m})$.

Proof. Rationality follows from (55). The analytic estimate is Theorem 13.3. Continued-fraction convergents satisfy $|q \log 3 - p \log 2| \ll q^{-1}$. $\square$

This is the strongest unconditional approximation statement obtained in this continuation. It gives arbitrary fixed contact order at the missing jet while using only the rational values forced by the original hypotheses.

14 Denominator and height audit

High-order contact is not by itself an irrationality proof. This section gives the exact reason, in a form suitable for deciding what kind of new lemma would be decisive.

14.1 Naive height of the rational extrapolants

Let (p, q) be a convergent pair and use

$$A = 2^p, \quad B = 3^q, \quad C = M^p, \quad D = N^q.$$

For fixed m , Formula (50) and the rational coefficients (51) give a completely explicit common denominator for $R_{m,x}(B/A)$. Since $p \asymp q$ along convergents, one obtains

$$\log \text{den}(R_{m,x}(B/A)) \leq C_{x,m} q \quad (68)$$

for an effective constant $C_{x,m}$, before any favorable cancellation is used. If m is allowed to vary, the crude product denominator gives

$$\log \text{den}(R_{m,x}(B/A)) = O_x(m^2 q + m^2 \log(m+1)). \quad (69)$$

These are upper bounds, not claims that the reduced denominator is always this large. The possible cancellation is precisely the unexplored arithmetic resource.

For fixed m , Corollary 13.5 supplies an error $O(q^{-2m})$, while (68) allows denominators of size $\exp(Cq)$. The reciprocal denominator scale is exponentially small, far below the polynomial analytic error. Consequently neither ordinary rational spacing nor an effective irrationality measure can yet lock the sequence.

14.2 Adjacent rational locking

Lemma 14.1 (Unit locking for two rational approximants). *Let P_j/Q_j and P_{j+1}/Q_{j+1} be reduced rationals with positive denominators. If*

$$\left| x - \frac{P_j}{Q_j} \right| + \left| x - \frac{P_{j+1}}{Q_{j+1}} \right| < \frac{1}{Q_j Q_{j+1}}, \quad (70)$$

then

$$\frac{P_j}{Q_j} = \frac{P_{j+1}}{Q_{j+1}}.$$

Proof. If the rationals were distinct, their distance would be at least $1/(Q_j Q_{j+1})$. The triangle inequality and (70) contradict this. $\square$

Closing criterion 14.2 (Richardson adjacent locking). *Choose $r_j = e^{h_j} \in \Gamma$ with $h_j \rightarrow 0$, orders $m_j \geq 1$, and write the reduced rational extrapolants as*

$$R_{m_j, x}(r_j) = \frac{P_j}{Q_j}.$$

Suppose explicit remainder bounds E_j are available with

$$\left| R_{m_j, x}(r_j) - x \right| \leq E_j$$

and

$$E_j + E_{j+1} < \frac{1}{Q_j Q_{j+1}} \quad (71)$$

for every sufficiently large j . Then $x \in \mathbb{Q}$, hence $x \in \mathbb{Z}$ under (1).

Proof. Lemma 14.1 makes the rational sequence eventually constant. Its real limit is x by Theorem 13.3, so x is that rational constant. $\square$

The criterion is elementary, exact, and stronger than any appeal to a generic irrationality measure. It identifies the desired denominator threshold without hiding constants.

14.3 Irrationality-measure locking

Because $x = \log M / \log 2$, lower bounds for nonzero linear forms in two logarithms give an effective irrationality measure: there exist computable $c > 0$, $\mu > 0$, and Q_0 such that

$$\left| x - \frac{P}{Q} \right| > cQ^{-\mu} \quad (72)$$

for all $Q \geq Q_0$ unless $x = P/Q$; see, for example, Baker–Wüstholz and the refinements surveyed in the unified report [4, 3].

Closing criterion 14.3 (Denominator-compression closure). *Fix m . Suppose there is a sequence $r_j = e^{h_j} \in \Gamma$, $h_j \rightarrow 0$, for which the reduced denominators Q_j of $R_{m,x}(r_j)$ satisfy*

$$C_{x,m}|h_j|^{2m} < cQ_j^{-\mu} \quad (73)$$

with $C_{x,m}$ an explicit remainder constant in Theorem 13.3. Then $x \in \mathbb{Z}$.

Proof. If the denominators Q_j are bounded along an infinite subsequence, the corresponding approximants lie in a fixed discrete subgroup and convergence already gives $x \in \mathbb{Q}$. Otherwise choose a term with $Q_j \geq Q_0$. The upper bound (60), together with (73), contradicts (72) unless $x = R_{m,x}(r_j) \in \mathbb{Q}$. Rational descent closes the problem. $\square$

A useful asymptotic version is the following. If

$$\log Q_j \leq \kappa_m \log \frac{1}{|h_j|} + O_m(1), \quad (74)$$

then (73) follows whenever $2m > \mu\kappa_m$. Thus arbitrary analytic order becomes decisive if the denominator exponent κ_m grows sublinearly in m . The current unreduced bounds do not have this form: along ordinary convergents, $\log Q_j$ is allowed to be proportional to $1/|h_j|$, not to $\log(1/|h_j|)$.

14.4 Why increasing m does not automatically fix the scale

One might try to choose $m = m(q) \rightarrow \infty$. Ignoring remainder constants, the analytic gain is roughly

$$|h|^{2m} \asymp q^{-2m} = \exp(-2m \log q),$$

whereas the crude common denominator in (69) has logarithm of order $m^2 q$. Even a more economical denominator assembled from the largest term typically carries an mq scale. Since $q \gg \log q$, the extra order does not cross the arithmetic spacing threshold. A proof must therefore obtain *cancellation in reduced denominators*, not merely more Taylor coefficients.

Caution 14.4. No lower bound for the reduced denominator of $R_{m,x}(r)$ is asserted here. Large exact gcds could radically reduce it, and finding such gcds is one of the best remaining opportunities. The audit says only that the presently visible, unreduced denominator is too large to combine with the analytic error.

15 New closing statements extracted from the analysis

The following statements are not reformulations of the conjecture by tautology; each isolates a concrete additional arithmetic property that would be sufficient.

Closing criterion 15.1 (Non-split extension closure). *Let A be any rational semisimple matrix with positive spectrum in Γ . If there exists $H \in M_n(\mathbb{Q})$ with nonzero class in $M_n(\mathbb{Q})/[A, M_n(\mathbb{Q})]$ such that $\mathcal{E}_A(H)^x$ is rational, then $x \in \mathbb{Z}$.*

This is Corollary 5.4, restated as a research interface.

Closing criterion 15.2 (Common-conductor confluence). *Suppose there are integral 3-smooth pairs (a_j, b_j) with $b_j/a_j \rightarrow 1$ such that the reduced normalized secants*

$$L_j = \frac{a_j}{a_j^x} \frac{b_j^x - a_j^x}{b_j - a_j} \in \mathbb{Q} \quad (75)$$

have bounded reduced denominators. Then $x \in \mathbb{Z}$.

Proof. The normalized secants converge to x . Bounded denominators put them in a fixed discrete subgroup of $\mathbb{Q}$, so they are eventually constant and their limit is rational. $\square$

Closing criterion 15.3 (Affine order preservation). *Suppose there is a sequence $q_j \in \Gamma$, $q_j \rightarrow 3$, and a fixed integer $D \geq 1$ such that*

$$D \frac{q_j^x - N}{q_j - 3} \in \mathbb{Z} \quad (76)$$

for every j . Then $x \in \mathbb{Z}$.

Proof. This is Proposition 11.1(3). $\square$

Closing criterion 15.4 (Sublinear Richardson denominator exponent). *Assume the effective irrationality measure (72). Suppose that for infinitely many orders m one can find sequences $h \rightarrow 0$ in $\log \Gamma$ such that*

$$\log \text{den}(R_{m,x}(e^h)) \leq (\kappa_m + o(1)) \log \frac{1}{|h|}, \quad \kappa_m = o(m).$$

Then $x \in \mathbb{Z}$.

Proof. Choose m with $2m > \mu\kappa_m$, then apply Criterion 14.3. $\square$

Closing criterion 15.5 (Rational derivative of the 3-smooth power homomorphism). *Regard ρ_x from (4) as a map on the dense subgroup $\Gamma \subset \mathbb{R}_{>0}$. If its logarithmic derivative at one rational point exists in an arithmetic sense that forces*

$$q \frac{\rho'_x(q)}{\rho_x(q)} \in \mathbb{Q} \quad (77)$$

for one $q \in \Gamma$, then $x \in \mathbb{Z}$.

Proof. Analytically, $q\rho'_x(q)/\rho_x(q) = x$. Rational descent finishes. $\square$

The phrase “in an arithmetic sense” is deliberately not left vague in the concrete targets above: bounded denominators, a preserved order, or a rational non-split block extension are three exact realizations.

16 Prioritized research program

16.1 Priority A: manufacture a nonzero extension class

The quotient theorem says exactly what must be built. Search for a canonical module or lattice on which the two known value maps act and whose self-extension class survives after applying x . Promising sources are:

1. affine semigroup actions combining dilation and translation;
2. modules of Kähler differentials or derivations of a suitable arithmetic semigroup algebra;
3. non-semisimple integral reductions followed by a characteristic-zero lifting theorem;
4. exact deformation rings in which the tangent direction is canonical rather than chosen externally.

The failure test is immediate: compute the class in $M_n(\mathbb{Q})/[A, M_n(\mathbb{Q})]$. If it is zero, the construction is only a disguised similarity.

16.2 Priority B: compute conductor defects on near-collisions

For convergents $2^p \approx 3^q$, compute

$$\gcd(3^q - 2^p, N^q - M^p) \tag{78}$$

and the symmetric analogues appearing in (50). The raw gap alone is not the statistic of interest; the ratio of the gap to this gcd is. Exact modular algorithms can handle much larger exponents than explicit integer expansion by using modular exponentiation, subresultants, and product trees.

A genuinely anomalous family with conductor defect polynomial in q would materially change the denominator audit. Conversely, a theorem proving exponential conductor defect would close off the entire confluence route and redirect effort toward extension manufacture.

16.3 Priority C: reduced-denominator Richardson tables

For each convergent (p, q) and modest order m , compute the rational number $R_{m,x}(3^q/2^p)$ symbolically in terms of M, N and factor its denominator as far as possible using only relations forced by $M = 2^x, N = 3^x$. The aim is not numerical evaluation of a nonexistent counterexample. It is to derive universal gcd identities or divisibility conditions in the polynomial ring $\mathbb{Z}[M, N]$ specialized to the power relation.

The coefficients (51) are explicit and the cancellation moments are exact, making this a good target for automated conjecture generation and formal checking.

16.4 Priority D: adelic confluence

A rational sequence can converge to an irrational real number because its denominators escape through finite places. The natural repair is adelic: seek $r_j \rightarrow 1$ in the real topology while controlling the secants in enough p -adic topologies that the product formula prevents escape. The challenge is simultaneous approximation in the exponent lattice:

$$a_j \log 2 + b_j \log 3 \rightarrow 0 \quad \text{and} \quad 2^{a_j} 3^{b_j} \rightarrow 1 \text{ in selected } \mathbb{Q}_p. \quad (79)$$

Congruence conditions on (a_j, b_j) can enforce the p -adic part; geometry of numbers can enforce the real part. What is missing is a transfer theorem for the powered values $M^{a_j} N^{b_j}$ strong enough to bound the secant denominators.

16.5 Priority E: affine S -unit rigidity

The family $C(q)$ in Section 11 asks for arithmetic regularity of the map $q \mapsto q^x$ at the rational point 3 when sampled on the S -unit group Γ . Classical S -unit equations show that exact additive patterns inside Γ are rare, but rarity alone is not enough. A useful theorem would control the heights of divided differences at an accumulation point, not merely count solutions to $u + v = 1$.

17 Lean formalization blueprint

The new algebraic core is unusually suitable for formalization. Most of it reduces to block matrix multiplication, finite spectral decompositions, and rational linear algebra. The analytic interface can be isolated.

17.1 Module 1: explicit 2–3 blocks

A first module can avoid general Fréchet derivatives entirely and formalize the concrete formula (25). Suggested declarations are:

```
-- Names and exact types are schematic, not compiled source.

theorem blockRpow_diag_two_three
  (x : Real) (M N a b c d : Rat)
  (hM : Real.rpow 2 x = M) (hN : Real.rpow 3 x = N) :
  blockMatrix (diag ![2, 3]) ![![a,b],[c,d]] ^ x =
    explicitPoweredBlock x M N a b c d := by
  ...

theorem rational_blockRpow_iff_offDiag_of_irrat
  (hx : Irrational x) (hM : (2 : Real) ^ x = M)
  (hN : (3 : Real) ^ x = N) :
  EntriesRational (block23 H ^ x) <=> H 0 0 = 0 /\ H 1 1 = 0 := by
  ...

theorem integer_of_two_rpow_integer_of_nonsplitBlock
  (h2 : (2 : Real) ^ x \in Set.range Int.cast)
  (h3 : (3 : Real) ^ x \in Set.range Int.cast)
  (hblock : EntriesRational (block23 diagDirection ^ x)) :
```

```
x \in Set.range Int.cast := by
...
```

Listing 1: Schematic Lean targets for the explicit block model.

The only analytic lemma needed is the power of a 2×2 Jordan block or, equivalently, the finite Taylor formula for a square-zero perturbation. The unified repository already contains a related `JordanFirstJet` endpoint, so this module should reuse rather than duplicate it.

17.2 Module 2: commutator splitting

The identities

$$M_n(\mathbb{Q}) = \mathcal{Z}(A) \oplus [A, M_n(\mathbb{Q})],$$

$$\mathcal{E}_A([A, Y]) = \begin{pmatrix} I & -Y \\ 0 & I \end{pmatrix} (A \oplus A) \begin{pmatrix} I & Y \\ 0 & I \end{pmatrix}$$

are finite algebra. A diagonal version should be formalized first; the spectral-projector version can follow.

```
theorem offDiag_eq_commutator_of_diag_distinct
  (lambda : Fin n -> Rat)
  (h : Pairwise fun i j => lambda i != lambda j)
  (H : Matrix (Fin n) (Fin n) Rat)
  (hdiag : forall i, H i i = 0) :
  exists Y, H = diagonal lambda * Y - Y * diagonal lambda := by
...

theorem commutatorExtension_isSimilar
  (A Y : Matrix (Fin n) (Fin n) Rat) :
  blockExtension A (A * Y - Y * A) =
    U (-Y) * blockDiag A A * U Y := by
...

theorem quotient_powerDerivative_eq_smul
  ... :
  quotientMap (powerFrechet x A H) =
    x . quotientMap (AinvTimesApow * H) := by
...
```

Listing 2: Schematic Lean targets for splitting and the quotient.

17.3 Module 3: the 2,3,4 Newton chain

The explicit matrix identity (30) can be proved by polynomial interpolation on three points, avoiding a general matrix-power library. The parity endpoint is elementary.

```
theorem newtonChain234_rpow
  (hM : (2 : Real) ^ x = M) (hN : (3 : Real) ^ x = N) :
  rpowMatrix T234 x =
    !! [M, N-M, (M^2-2*N*M)/2;
        0, N, M^2-N;
        0, 0, M^2] := by
```

```

...

theorem even_Msq_sub_twoN_add_M (M N : Int) :
  Even (M^2 - 2*N + M) := by
  omega_or_ring

theorem eigenlattice_index_T234 :
  abs (Matrix.det primitiveEigenvectors234) = 2 := by
  norm_num

```

Listing 3: Schematic Lean targets for the integral Newton chain.

17.4 Module 4: Richardson coefficients

This is pure finite combinatorics over $\mathbb{Q}$.

```

def richardsonCoeff (m k : Nat) : Rat :=
  (-1)^(k-1) * (2 * (m.factorial)^2) /
  ((m-k).factorial * (m+k).factorial)

theorem richardsonCoeffs_sum (hm : 1 <= m) :
  sum (fun k => richardsonCoeff m k) (Icc 1 m) = 1 := by
  ...

theorem richardsonCoeffs_moment
  (hj : 1 <= j) (hjm : j < m) :
  sum (fun k => richardsonCoeff m k * k^(2*j)) (Icc 1 m) = 0 := by
  ...

theorem richardsonCoeffs_firstSurviving :
  sum (fun k => richardsonCoeff m k * k^(2*m)) (Icc 1 m) =
    (-1)^(m-1) * (m.factorial)^2 := by
  ...

```

Listing 4: Schematic Lean targets for Richardson cancellation.

The cleanest proof uses Lagrange interpolation at the nodes k^2 . If existing Mathlib interpolation infrastructure is awkward, a direct rational-function identity after clearing factorials is likely shorter.

17.5 Module 5: analytic remainder interface

The analytic theorem can be stated independently of the arithmetic application:

```

def symmetricSecant (x h : Real) : Real := sinh (x*h) / sinh h

theorem symmetricSecant_evenAnalytic :
  HasFPowerSeriesAt (fun h => symmetricSecant x h) evenSeries 0 := by
  ...

theorem richardsonSecant_isLittleO
  (m : Nat) (hm : 1 <= m) :
  (fun h => richardsonCombination m x h - x) =o[nhds 0]
  (fun h => h^(2*m-1)) := by
  ...

```

Listing 5: Schematic analytic interface.

For the arithmetic report, a weaker explicit bound on a compact interval is sufficient and may be easier to formalize than the exact power series.

17.6 Dependency order and trust boundary

A practical build order is:

1. Richardson coefficient identities;
2. explicit matrix and similarity identities;
3. the 2–3 rationality test;
4. the 2, 3, 4 chain and parity;
5. diagonal finite-dimensional derivative formula;
6. general spectral-projector formulation;
7. analytic Richardson remainder;
8. final conditional closing criteria.

The first four stages require no sophisticated real matrix power API. They can therefore be checked quickly and provide a reliable algebraic core even if the general holomorphic functional-calculus layer takes longer.

18 Conclusions

The strongest new result is not another sufficient condition of the form “if one somehow knows a derivative, then x is rational.” It is a complete theorem about *which* first-order extensions can remain rational under a hypothetical irrational exponent: precisely those that were already split. This gives an exact quotient on which power acts as x times a rational automorphism. Any successful finite-dimensional route must put a nonzero rational vector into that quotient.

The toric constructions exhaust a large and tempting class of candidates and provably miss the quotient. Integral lattice gluing and repeated tensor eigenvalues do not help by themselves; the explicit matrix T and its tensor square demonstrate the failure concretely. Singular conjugation does recover the Jordan jet in the limit, which explains why secant methods feel so close, but the conjugators and normalizing denominators become unbounded.

The symmetric/Richardson hierarchy is a real quantitative improvement: it converts the prior first-order rational secants into rational approximations with arbitrary fixed even contact order. It also makes the remaining obstruction more unforgivingly precise. The missing lemma must compress reduced denominators, preserve a nontrivial conductor, or supply an adelic compactness mechanism. More Taylor expansion alone will not suffice.

The most promising next move is therefore two-pronged:

1. algebraically, search for a canonical non-split extension arising from the additive relation $3 = 2 + 1$ or from a derivation of an arithmetic semigroup algebra;

2. computationally, measure the common conductor and reduced Richardson denominators on very large 2^p-3^q near-collisions, looking for exact gcd structure rather than merely small real gaps.

These targets are narrow enough for formalization and exact computation, yet genuinely outside the semisimple barrier already mapped by the unified report. They do not prove the conjecture, but they materially reduce the space in which a proof can hide.

A Quantitative Richardson bounds

The main text uses $O_{x,m}(h^{2m})$, which is sufficient for the structural conclusions. For unit-locking experiments one wants an explicit error bound.

Proposition A.1 (A Cauchy remainder bound). *Fix $x \in \mathbb{R}$, $m \geq 1$, and $0 < \rho < \pi$. Let*

$$F_x(z) = \begin{cases} \sinh(xz)/\sinh z, & z \neq 0, \\ x, & z = 0, \end{cases} \quad (80)$$

and put

$$M_{x,\rho} = \max_{|z|=\rho} |F_x(z)|, \quad C_m = \sum_{k=1}^m |c_{m,k}|. \quad (81)$$

If $m|h| < \rho$, then

$$|R_{m,x}(e^h) - x| \leq C_m M_{x,\rho} \frac{(m|h|/\rho)^{2m}}{1 - (m|h|/\rho)^2}. \quad (82)$$

Proof. The function F_x is even and holomorphic on $|z| < \pi$ because the zero of $\sinh z$ at zero is removable and the next zeros are at $\pm\pi i$. Write $F_x(z) = x + \sum_{j \geq 1} a_j z^{2j}$. Cauchy's estimate gives $|a_j| \leq M_{x,\rho} \rho^{-2j}$. Moment cancellation removes all $j < m$, so

$$\begin{aligned} |R_{m,x}(e^h) - x| &\leq \sum_{k=1}^m |c_{m,k}| \sum_{j \geq m} |a_j| |kh|^{2j} \\ &\leq C_m M_{x,\rho} \sum_{j \geq m} (m|h|/\rho)^{2j}, \end{aligned}$$

which is the displayed geometric sum. □

For reference, the first coefficient vectors and their ℓ^1 norms are:

m	$(c_{m,1}, \dots, c_{m,m})$	C_m
1	1	1
2	4/3, -1/3	5/3
3	3/2, -3/5, 1/10	11/5
4	8/5, -4/5, 8/35, -1/35	93/35
5	5/3, -20/21, 5/14, -5/63, 1/126	193/63
6	12/7, -15/14, 10/21, -1/7, 2/77, -1/462	793/231

The growth of C_m is modest compared with the factorial in the first surviving moment, but this does not change the denominator-scale diagnosis: the dominant arithmetic cost comes from the sampled rational values, not from the Richardson coefficients.

B Further structural consequences of the extension theorem

B.1 Multiple eigenvalues

Theorem 5.3 allows each spectral projector P_i to have arbitrary rank. On the block $P_i M_n(\mathbb{Q}) P_i$, the derivative is multiplication by the scalar $x\lambda_i^{x-1}$. Thus every nonzero rational matrix coefficient within that block detects x . The quotient action is not one-dimensional; it is a direct sum of full matrix spaces, but the same scalar x appears on every summand after rational rescaling.

In particular, extra multiplicity does not weaken the detector. What weakens it is semisimplicity: a torus construction supplies no nonzero class in any of those matrix spaces.

B.2 Higher triangular blocks

For an invertible A and a nilpotent N commuting with A and satisfying $N^{r+1} = 0$, one has the terminating formula

$$(A + N)^x = \sum_{j=0}^r \binom{x}{j} A^{x-j} N^j. \quad (83)$$

If A^x is rational and the coefficient of any nonzero N^j can be shown rational, then one obtains arithmetic information about the falling factorial $x(x-1)\cdots(x-j+1)$. The first coefficient already suffices, so higher jets are not needed for logical closure. They may nevertheless be easier to manufacture as limits of symmetric powers, as Corollary 10.3 shows.

B.3 Derivations of the value algebra

Let R be the rational algebra generated by a semisimple A . A self-extension direction commuting with A can be interpreted as adjoining a square-zero element εH and asking for a lift of the map

$$A \mapsto A^x$$

to $R[\varepsilon]/(\varepsilon^2)$. The lift necessarily sends

$$A + \varepsilon H \mapsto A^x + \varepsilon x A^{x-1} H.$$

Thus the missing datum is a derivation-level lift of the value homomorphism. Semisimple torus representations determine the map on reduced algebras but provide no canonical square-zero thickening. This algebraic language may be a more natural starting point than matrices for constructing the required non-split class.

C Symbolic verification script

The following short SymPy script independently checks the identities that are most vulnerable to transcription errors: the centralizer/commutator block formula in the 2–3 model, the 2, 3, 4 Newton-chain matrix, the first terms of the symmetric-secant expansion, the Richardson moment identities through order 10, and the sign in the split-to-Jordan degeneration.

```

1 from __future__ import annotations
2
3 from math import factorial
4 import sympy as s
5
6 # Symbols.
7 x, h = s.symbols("x h")
8 M, N = s.symbols("M N")
9 a, b, c, d = s.symbols("a b c d")
10
11 # 1. The 2--3 Frechet derivative formula from spectral blocks.
12 A = s.diag(2, 3)
13 H = s.Matrix([[a, b], [c, d]])
14 P2 = s.diag(1, 0)
15 P3 = s.diag(0, 1)
16 D = (
17     x * M / 2 * (P2 * H * P2)
18     + x * N / 3 * (P3 * H * P3)
19     + (N - M) * (P2 * H * P3 + P3 * H * P2)
20 )
21 assert D == s.Matrix([[x*M*a/2, (N-M)*b],
22                       [(N-M)*c, x*N*d/3]])
23
24 # 2. The 2,3,4 interpolation polynomial and its value at T.
25 z = s.symbols("z")
26 p = s.interpolate([(2, M), (3, N), (4, M**2)], z)
27 T = s.Matrix([[2, 1, 0], [0, 3, 1], [0, 0, 4]])
28 I = s.eye(3)
29 pT = s.expand(p).coeff(z, 0) * I
30 pT += s.expand(p).coeff(z, 1) * T
31 pT += s.expand(p).coeff(z, 2) * (T*T)
32 expected = s.Matrix([
33     [M, N-M, (M**2-2*N+M)/2],
34     [0, N, M**2-N],
35     [0, 0, M**2],
36 ])
37 assert s.simplify(pT - expected) == s.zeros(3)
38
39 # 3. Symmetric-secant Taylor series.
40 series = s.series(s.sinh(x*h) / s.sinh(h), h, 0, 8)
41 expected_series = (
42     x + x*(x**2-1)*h**2/6
43     + x*(3*x**4-10*x**2+7)*h**4/360
44     + x*(3*x**6-21*x**4+49*x**2-31)*h**6/15120
45 )
46 assert s.expand(series.removeO() - expected_series) == 0
47
48 # 4. Richardson coefficients and exact moments.
49 def coeff(m: int, k: int) -> s.Rational:
50     return s.Rational(
51         (-1)**(k-1) * 2 * factorial(m)**2,

```

```

52     factorial(m-k) * factorial(m+k),
53 )
54
55 for m in range(1, 11):
56     cs = [coeff(m, k) for k in range(1, m+1)]
57     assert sum(cs) == 1
58     for j in range(1, m):
59         assert sum(cs[k-1] * k**(2*j)
60                     for k in range(1, m+1)) == 0
61     assert sum(cs[k-1] * k**(2*m)
62               for k in range(1, m+1)) == (-1)**(m-1) * factorial(m)**2
63
64 # 5. Correct sign in the split-to-Jordan conjugation.
65 a0, r, k = s.symbols("a0 r k", nonzero=True)
66 b0 = a0*r
67 S = s.Matrix([[1, -k], [0, 1]])
68 B = s.simplify(S.inv() * s.diag(a0, b0) * S)
69 expected_B = s.Matrix([[a0, k*(b0-a0)], [0, b0]])
70 assert s.simplify(B - expected_B) == s.zeros(2)
71
72 print("all exact audits passed")

```

Listing 6: Exact symbolic audit used for this report.

The script uses exact rational and symbolic arithmetic. It is a transcription check, not a substitute for the proofs in the main text and not a proof-assistant certificate.

D Status and dependency matrix

Table 2: Status of the principal statements.

Statement	Status	Dependencies and caveats
Block derivative identity, Theorem 4.1	[proved here]	Standard holomorphic functional calculus; contour proof included
Semisimple divided-difference formula, Theorem 4.2	[proved here]	Finite spectral decomposition
Exact rationality locus, Theorem 5.3	[proved here]	Requires rational spectral projectors and rational powered eigenvalues; no transcendence conjecture
Extension quotient action, Theorem 6.1	[proved here]	Basis-free consequence of the rationality-locus decomposition
Toric-functor no-go, Theorem 8.1	[proved here]	Standard semisimplicity of split-torus representations in characteristic zero
2, 3, 4 Newton chain, Theorem 9.1	[proved here]	Exact interpolation and a parity check; symbolically audited

Statement	Status	Dependencies and caveats
Split-to-Jordan degeneration, Theorem 10.1	[proved here]	Uses unbounded integral conjugators; gives a limit, not arithmetic compactness
Symmetric secant expansion, Theorem 12.2	[proved here]	Elementary power series; symbolically audited
Richardson moment identities and $O(h^{2m})$ theorem	[proved here]	Lagrange interpolation at nodes k^2 ; symbolically audited through $m = 10$
Explicit Cauchy remainder, Proposition A.1	[proved here]	Valid for $m h < \rho < \pi$
Denominator-compression and locking criteria	[proved here]	Conditional statements; the required denominator estimates remain open
Full Alaoglu–Erdős conjecture	[open interface]	Not proved here or in the attached manuscript

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